

Kawasaki's disease

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Kawasaki's disease can cause cardiac changes. Such changes have implications for dental treatment.

Kawasaki's disease or mucocutaneous lymph node syndrome (MCLS) is an acute febrile disease complex of unknown etiology that occurs most commonly in children under 5 years of age. Based on 50 cases, MCLS was first reported in Japan in 1967.¹ Since that time, nearly 18,000 cases have been recorded in that country, where it is considered to be of epidemic proportion. The frequency of reported cases is increasing in the United States and around the world.

The most critical feature of MCLS is cardiovascular changes, which may include aneurysm formation, coronary thromboarteritis, mitral valve insufficiency, and myocardial ischemia.² These changes influence the medical-dental management of a patient with history of this disease. The purpose of this paper is to acquaint dentists with this syndrome, its possible sequelae, and the implications for dental treatment.

Clinical features

Diagnostic criteria

Diagnosis of Kawasaki's disease is based solely on clinical evaluation. Six diagnostic criteria are outlined, five of which must be met to confirm the diagnosis.

—Fever (101 F - 104 F) lasting from one to two weeks and not responding to antibiotics.

—Bilateral congestion of ocular conjunctivae.

—Changes of lips and oral cavity: dryness, redness, and fissuring of lips (Fig 1); protuberance of tongue papillae (strawberry tongue-Fig 2); and diffuse reddening of oral and pharyngeal mucosa.

—Changes in the peripheral extremities: reddening of palms and soles and indurative edema, which starts on the third to fifth days and disappears in less than a week; and membranous desquamation from fingertips during second week of illness (Fig 3).

—Polymorphous exanthema of body trunk without vesicles or crusts on the fifth to seventh days.

—Acute nonpurulent swelling of cervical lymph nodes to 1.5 cm or more in diameter.

A transverse furrow of the nails is usually seen one to two months after onset of the disease.³⁻⁵ Other significant clinical features occasionally noted during the acute stage of the illness are diarrhea, arthralgia and arthritis, meatitis, tympanitis, photo-

phobia, meningeal irritation, and pneumonia.²

The cause of Kawasaki's disease is unknown, and there is no single test or battery of tests that is diagnostic. However, several laboratory results are frequently associated with Kawasaki's disease: proteinuria and increased numbers of leukocytes in urine sediment, leukocytosis with shift to the left, slight anemia, increased ESR (erythrocyte sedimentation rate), positive CRP (C-reactive protein), increased serum α_2 globulin concentrations, and negative ASO titer.^{2,5}

Cardiovascular involvement

The most serious sequelae of Kawasaki's disease are the coronary arterial changes, most notably the occurrence of "bead-like" aneurysms and coronary thromboarteritis. The fatality rate is 1% to 2%, and most deaths occur suddenly during the first three to four weeks of the disease.⁶ Death is usually caused by the rupture of an aneurysm and extensive thrombosis, with possible evidence of myocardial infarction or ischemia.^{7,8}

Patients at highest risk for sudden death are infant males with an extremely elevated ESR and prolonged fever and rash.⁷ Other cardiac symptoms include shock, gallop rhythm, distant heart sounds, apical systolic heart murmur, arrhythmia, and electrocardiographic changes.^{6,9}

Abnormal angiograms, with changes ranging from mild artery tortuosity and



Fig 1 ■ Dryness, redness, and fissuring of lips.



Fig 3 ■ Membraneous desquamation from fingertips.



Fig 2 ■ Red "strawberry" tongue.

dilatation to aneurysm formation and significant stenosis, occur in approximately 30% to 60% of Japanese patients.² In another study, 69 of 73 cases had some type of electrocardiographic changes.¹⁰ There are also reports of mitral valve insufficiency caused by papillary muscle dysfunction, as well as cardiomegaly and hypertrophy.^{2,11-13} The coronary artery disease associated with MLNs has, in some cases, regressed as documented by serial angiographic studies. In other patients, the disease process progressed to ischemic heart disease, and sudden death has been reported up to eight years after onset.^{6,14,15} In any case, long-term followup with serial cardiac evaluation is indicated.¹⁶ Noninvasive evaluation of cardiac involvement has been performed with echocardiography.^{17,18}

Management of Kawasaki's disease falls into two areas, pharmacotherapeutic and surgical. Corticosteroids, antibiotics, and anticoagulants have been the major agents used for drug therapy. Antibiotics have generally been ineffective,² and corticosteroids may adversely affect the coronary lesions. Kato and associates report that aspirin, with its anti-inflammatory and antithrombotic properties, may be effective in preventing sudden death caused by coronary thrombosis.¹⁴ Other anticoagulants, such as coumarin and heparin, have been

used in patients with evidence of occlusive vascular disease, especially in the limbs.¹¹

Surgical management may include aneurysmorrhaphy, thrombectomy for aneurysm, aortocoronary bypass grafting, ventricular aneurysmectomy, and mitral valve replacement.¹²

Epidemiology

The Center for Disease Control has been analyzing data on Kawasaki's disease in the United States since 1975. By 1978, more than 250 suspected cases¹¹ had been reported in 31 states and Washington, DC. The male to female ratio was 1.52:1 and no geographic or seasonal trends were apparent. Fifty-seven percent of patients were white, 18% black, and 25% oriental. The mean age of the confirmed cases was 3.8 years, which is higher than the age of peak incidence in Japan (18 months).¹⁹⁻²¹ Although the majority of cases are found in children, there have been cases reported in adults as old as 27.²²

Discussion

The most obvious of concerns to the dentist treating a patient with a history of Kawasaki's disease are the cardiac manifestations. The practitioner must

be aware that the patient may have a mitral valve insufficiency, which necessitates prophylaxis for subacute bacterial endocarditis (SBE) (Table 1).^{23,24}

With possible ischemia or infarction or both of the myocardium, the dentist must also be cognizant of the amount of stress caused in the child by the dental operator. Increasing stress may cause endogenous epinephrine release, which increases cardiac rate, force of contraction, and output; as a result, the work of the heart and its oxygen consumption increase and cardiac efficiency (work done relative to oxygen consumption) is lessened.²⁵ In a patient with ischemia, the compromised coronary blood flow may not be able to keep pace with the increased oxygen requirement and could result in further cardiac damage.

Dentists who treat children have long recognized the importance of stress management from a psychological standpoint. This particular disease complex, however, requires stress management from a physiological standpoint as well, because Kawasaki's disease may put the practitioner in the unusual position of managing an ischemic myocardium in a child.

Suggestions for stress management and inhibition of endogenous epinephrine release during dental treatment would include the following: good behavior management technique, short appointments, maintaining good ventilation, nitrous-oxide sedation, appropriate drug sedation (respiratory depressants are contraindicated), and general anesthesia.

These measures would certainly be

influenced by other factors relative to the particular situation, such as the patient's age and behavior, amount of treatment to be completed, and consultation with the patient's physician.

Another concern when treating a patient with an ischemic myocardium is the use of vasoconstrictors in local anesthetics. Allen and associates found that the 18 µg of epinephrine found in one carpule of 2% lidocaine with 1:100,000 epinephrine produced cardiovascular changes within five minutes of injection and that these actions were prolonged.²⁶ These researchers also noted linear increases of cardiac rate, output, and stroke volume with increased amounts of epinephrine. These findings support the contention that even small amounts of epinephrine can effect cardiac changes. Therefore, conservatism in epinephrine administration is essential in cardiac patients of this type.

Table 2 lists several vasoconstrictors used in anesthetics and the maximum adult dose for cardiac patients as given by Monheim.²⁷ The maximum pediatric dose for the average 3-year-old patient was calculated by using Clark's rule, which states: child's dose = weight (pounds) / 150 × adult dose. The fiftieth percentile in body weight for a 3-year-old is approximately 32.5 lb,²⁸ and this weight was used in making the calculations in Table 2.

Because the maximum allowable dose of vasoconstrictor for a 3-year-old child is so low (usually approximately a half Carpule), it may be advisable to use a local anesthetic without vasoconstrictor (for example, 3% mepivacaine). Although the duration of anesthesia will be lessened without use of a vasoconstrictor, the average working time of 3% mepivacaine is approximately 20 to 40 minutes, which is adequate for most procedures. As mentioned, short appointments may also reduce stress in children, which would affect endogenous catecholamine release.

Other factors that affect the patient with cardiovascular disease are failure to aspirate when local anesthesia is administered; too rapid administration of anesthetic; enhanced absorption of the pressor agent from inflamed areas; neglect in premedicating a child to reduce the amount of endogenously released epinephrine; and inability to achieve complete anesthesia, which results in higher levels of the en-

Table 1 ■ Antibiotic prophylaxis of children with problems affecting the heart.*

Regimen A: Penicillin

Parenteral-oral combined

1. Aqueous crystalline penicillin G (30,000 U/kg IM) mixed with procaine penicillin G (600,000 UIM) 30 to 60 minutes before the dental procedure.
2. Then penicillin V, 500 mg, given orally every six hours for eight doses.
3. In children less than 60 pounds, penicillin V, 250 mg, given orally every six hours for eight doses.

Oral only

1. Penicillin V, 2 gm orally 30 to 60 minutes before the dental procedure, then 500 mg every six hours for eight doses.
2. In children less than 60 pounds, penicillin V, 1 gm orally 30 to 60 minutes before the dental procedure, then 250 mg every six hours for eight doses.

Regimen B: Penicillin plus streptomycin

1. Aqueous crystalline penicillin G (30,000 U/kg IM) mixed with procaine penicillin G (600,000 UIM) plus streptomycin (20 mg/kg IM), 30 to 60 minutes before the dental procedure.
2. Followed by oral penicillin V, 500 mg every six hours for eight doses.
3. In children less than 60 pounds, oral penicillin V, 250 mg every six hours for eight doses.

Patients allergic to penicillin

1. Parenteral vancomycin plus erythromycin, vancomycin (20 mg/kg IV over 30 to 60 minutes) plus oral erythromycin (10 mg/kg every six hours for eight doses).
2. Oral erythromycin: 20 mg/kg orally 1½ to 2 hours before the dental procedure and 10 mg/kg every six hours for eight doses.

*Rheumatic or other valvular heart disease, congenital heart disease (ventricular septal defect, tetralogy of Fallot, aortic stenosis, pulmonic stenosis, complex cyanotic heart disease, patent ductus arteriosus, or systemic to pulmonary artery shunts), idiopathic hypertrophic subaortic stenosis, mitral valve prolapse syndrome with mitral insufficiency, and cardiac valvular prostheses.

Note: for more information, consult references 23, 24.

Table 2 ■ Vasoconstrictor dosage in ischemic heart patients.

Vasoconstrictor	Maximum adult dose	No. Carpules	Maximum dose for 32.5-lb child	No. Carpules
Epinephrine 1:100,000 (0.01 mg/ml)	0.04 mg	2.22	0.0087	0.48
Levarterenol 1:30,000 (0.033 mg/ml)	0.14 mg	2.36	0.030	0.51
Corbasil 1:10,000 (0.1 mg/ml)	0.4 mg	2.22	0.0087	0.48

dogenous hormone caused by apprehension.²⁹

Parenthetically, some researchers have found that local anesthetic solutions without vasoconstrictors cause less injection pain than those anesthetics with a vasoconstrictor. Oikarinen and others also stated that a low pH anesthetic more frequently caused pain than a higher pH anesthetic.^{30,31} These results would again favor the use of 3% mepivacaine, which has a relatively higher pH than other anesthetics and is without vasoconstrictor.

If the practitioner uses 3% mepivacaine or any other local anesthetic without vasoconstrictor, it must be recognized that this mode of treatment is not without limitations. First, anesthetics without a vasoconstrictor may prolong the time of onset of anesthesia. Furthermore, with 3% mepivacaine, the maximum allowable dose is 3 mg/lb. For a 3-year-old child (using 32.5 lb as an average), this would translate into a maximum dose of 97.5 mg or only 1.81 carpules. Therefore, the dentist must be aware of, and be careful with, local anesthetic dosage.

Summary

Kawasaki's disease (mucocutaneous lymph node syndrome) is an acute febrile disease complex of unknown etiology that occurs most commonly in children younger than 5 years. Mitral valve insufficiency, aneurysm and thrombosis of the coronary vasculature, and myocardial ischemia are possible sequelae of the disease process. Precautions for dental treatment are described.

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Orofacial pain of cardiac origin

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Dentists should not assume that all orofacial pain is of dental origin. A case of orofacial pain caused by impending myocardial infarction is described.

Treatment of orofacial pain is one of the most common and perplexing problems that confronts the dentist. Most often the diagnosis is easily made through X rays and other diagnostic aids. Sometimes our most used aids provide little in the way of finding the cause of such pain.

We, as dentists, usually assume that orofacial pain is of dental origin. As we

become more experienced and see greater numbers of patients, we find many other causes of such pain. Some causes of orofacial pain, other than dental origin, are tic douloureux, various vascular disorders, giant cell arteritis¹, and cardiovascular disease.² In this case report of cardiovascular disease, the only initial symptom was orofacial pain.

Report of case

On May 3, a 41-year-old white woman was awakened at 6 AM by acute pain in the left jaw and went to the emergency room. The emergency room personnel found her in no acute distress other than a toothache, and she was referred to the oral surgery clinic for treatment.

A panoramic X ray was taken. When the patient was questioned, a possible diagnosis was made of periodontitis in the area of the mandibular right first and second premolars. Her medical history was essentially noncontributory. The patient was given a mandibular block anesthetic to relieve the pain. After the anesthetic took effect, a more thorough gingival examination, as well as a gingival curettage, could be done. However, 15 minutes after the administration of the local anesthetic, the patient continued to have facial pain. The patient then became diaphoretic and had crushing substernal chest pain and pain radiating down the medial aspect of the left arm. Until that time, the patient had had only oral pain and no other symptoms. She was then taken back to the emergency room with a possible diagnosis of myocardial infarction.